Reinforcement Learning Laboratory

Lab Assignment 3: GridWorld – Policy Evaluation and Value Iteration

Course Outcomes: CO2, CO3, CO4

Objective

Formulate a GridWorld environment as a Reinforcement Learning problem, implement Policy Evaluation and Value Iteration, and compare different policies.

Instructions

- Study the GridWorld environment and identify states, actions, rewards, and terminal states.
- Implement Policy Evaluation for a given policy.
- Implement Value Iteration to determine the optimal policy.
- Compare paths generated by random and optimal policies.
- Record observations and answer all analysis questions.

Task 1: Formulating the GridWorld Environment

RL Component	Description
States	
Actions	
Reward Structure	
Terminal State(s)	
Goal of the Agent	

Task 2: Policy Evaluation

Iteration	Maximum Change (Δ)	Convergence Status
1		
2		
3		
4		
5		

Task 3: Value Iteration and Optimal Policy Generation

State	Optimal Action	State Value
S1		
S2		
S3		
S4		
S5		

Task 4: Optimal Path Analysis

Policy Type	Path Length	Total Reward	Goal Reached (Yes/No)

Random Policy			
Evaluated Policy			
Optimal Policy			

Analysis Questions

What is a state in GridWorld?

What actions can the agent perform?

What is the purpose of Policy Evaluation?

How does Value Iteration differ from Policy Evaluation?

What is the Bellman Optimality Equation?

Which policy reached the goal using the fewest steps?

Why is the optimal policy superior to a random policy?

How would obstacles affect the optimal path?